

Dr Cole Robertson

Researcher in Cognitive Neuroscience & AI

Interdisciplinary cognitive scientist and AI researcher specializing in the mechanistic study of reasoning, representation, and goal-directed behavior in human and machine cognition. Grounded in the experimental study of human intelligence, my work adapts cognitive-neuroscience methods — functional localization, targeted ablation, and interventional experiment — to develop the science of interpretable AI evaluation, differentiating performance from understanding, reasoning, and goals. I apply these techniques to frontier AI systems to diagnose failure modes and guide the development of safer, more interpretable, and better aligned artificial intelligence. This path has been circuitous: trained first in the humanities and evolutionary anthropology, then in cognitive science, before co-founding two AI companies. Having exited Ogma Therapy via trade sale, I am now returning to fundamental research, convinced that I can make a deeper contribution upstream, on the mechanistic understanding that safe deployment depends on. I am extending this programme to multi-agent safety: detecting collective goal drift in populations of generalizing agents, and asking how training environments select for prosocial behavior. My ambition is to close the loop between the study of human intelligence and the science of AI, so that a mature toolkit for investigating cognition can be applied to the systems whose internal workings we most urgently need to understand.

EDUCATION

Technical AI Safety Course

BlueDot Impact

Accepted 2026

PhD in Cognitive Science and Linguistics

Radboud University

November 2021

Dissertation: The future is a thing of possibility: Cross-linguistic differences in modal future time reference affect (risky) intertemporal decisions. Supervisors: Prof Robin Dunbar (Oxford); Prof Asifa Majid (Radboud); Dr Seán Roberts (Max Planck). Nominated: Glushko Dissertation Prize, Cognitive Science Society.

MSc in Cognitive and Evolutionary Anthropology

University of Oxford

September 2014

Supervisor: Prof Harvey Whitehouse. Awards: Best Use of Quantitative Methods.

BA in Western Society and Culture

Concordia University

May 2011

Awards: Great Distinction.

RESEARCH EXPERIENCE

Visiting Researcher (Remote)

Emory University, USA

April 2024 – Present

- Since co-founding Ogma Therapy, I have maintained a collaboration with Prof Wolff, continuing to develop cognitive-neuroscience techniques to evaluate cognitive processes in LLMs.
- Conducted functional localization and ablation experiments to identify latent representations involved in mechanical reasoning in foundation models (work in prep).
- Completed the final study of the mechanical-reasoning paper and wrote it up unpaid alongside the Ogma CTO role; the white-box extension is self-funded from consulting income.

Postdoctoral Researcher

Emory University, USA

October 2022 – April 2024

- Working with Prof Philip Wolff, I developed novel approaches to evaluating multimodal foundation models on mechanical reasoning tasks (Superintelligence Conference 2025; [arXiv:2507.15521](https://arxiv.org/abs/2507.15521)).
- Conducted independent literature review to identify research opportunities by connecting the literature on human mental models and AI world models.
- Designed original stimuli and experiments: 810 novel pulley-system problems encoded in TikZ and withheld from the public web to keep the test out of

CONTACT

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- [Google Scholar](#)

SKILLS

Interpretability & neuro-inspired methods

- Functional localization — identifying units and subnetworks selectively engaged by a task
- Targeted ablation and lesioning, with random-mask controls
- Representational probing; linear probes on hidden states
- Activation steering and causal intervention
- Feature attribution — Integrated Gradients (Captum), SHAP, LIME
- Surrogate models; local versus global explanation; attribution stability across reruns
- Cognitive-science paradigms applied to models — mental models, reasoning tasks, decision-making under risk and uncertainty

Evaluation & robustness

- Controlled out-of-distribution paradigms; procedural stimulus generation to prevent contamination
- Stress testing with adversarial and edge-case inputs
- Domain-shift and distribution-shift analysis
- Classification metrics — accuracy, F_1 , ROC-AUC, PR-AUC; regression metrics — RMSE, MAE, R^2
- Calibration and reliability
- Error-analysis pipelines — qualitative, quantitative, slice-based; performance on critical subgroups (e.g., clinical populations)

Machine learning

- Core methods — regression (linear, logistic, regularized), tree-based models and ensembles, SVMs, k-NN, clustering
- Dimensionality reduction — PCA, t-SNE, UMAP
- Deep learning — sequence models, transformers, representation learning, regularization, optimization
- Model selection and validation — cross-validation schemes, nested CV, hyperparameter search, leakage prevention
- Stacked ensembles and

distribution; 4,050 trials on each of four frontier models. Models attended to causally relevant variables but estimated mechanical advantage with a pulley-counting heuristic, and fell to chance ($F_1 = 0.46$) when success required representing structural connectivity.

- Developed techniques which leverage explainable AI to identify linguistic biomarkers of amyloidosis pathology (npj Dementia, [here](#)): fine-tuned DistilRoBERTa on connected speech reached 92% accuracy over 10×10 -fold cross-validation, ten points above prior work, with LIME surrogate models 97% faithful to the classifier.

Research Assistant

September 2017 – September 2018

Brunel University, UK

- During my PhD, I interned with Dr James Carney, conducting original experimental design, literature review, and data analyses of unstructured social media data.
- Published first-author paper in Current Research in Behavioral Sciences, as well as collaborated with Dr Carney to co-author numerous other papers.

Visiting Researcher

2016 – 2017

The Max Planck Institute for Psycholinguistics

- Conducting independent research towards the completion of my PhD.

Research Assistant

September 2014 – September 2015

University of Oxford, UK

- Assisting postdocs and PhD students in Robin Dunbar's Social and Evolutionary Neuroscience Research Group in conducting original research for publication in academic journals and presentation at conferences.
- Published a first-author paper in Ethology.

TECHNICAL LEADERSHIP AND APPLIED AI

Independent Consultant and Fractional CTO

2026 – Present

UK

- Fractional CTO, Allie: preparing a React Native app with a Supabase backend for production release, in women's reproductive health.
- Speech-recognition R&D on disordered child speech for Ogma's acquirer.
- Fractional CTO to early-stage companies in demand forecasting and payments.

Chief Technology Officer

April 2024 – April 2026

Ogma Therapy, UK

- Co-founded Ogma Therapy to use AI/ML to help children with speech, language, and communication needs access speech therapy.
- Led scientists, data collectors, and product teams in training child-focused phonetic automatic speech recognition technology: fine-tuned wav2vec2 XLS-R CTC phoneme models (PyTorch, AWS SageMaker) with a bespoke 38-token IPA vocabulary, ensembled with autoregressive encoder-decoder models whose fixed-vocabulary decoding otherwise overwrites the misarticulations of clinical interest; phoneme error rates below 10%.
- Implemented quality-assured annotation pipelines to label data to train large acoustic models on goodness of pronunciation estimation (Prodigy, inter-annotator agreement), since no ground truth existed for phonetic transcription of disordered child speech.
- Automated therapy interventions using fine-tuning, RAG, and prompt-engineering.
- Implemented AI-safety and child safeguarding layers to ensure safe deployment of automated therapy agents.
- Raised £700k+ through equity financing; led negotiations to sell Ogma Therapy via a trade sale.

Chief Technology Officer

September 2018 – October 2022

Texture AI, UK

- During my PhD, I co-founded Texture AI to use NLP to tackle difficult communication problems faced by governments and third-sector organizations.

meta-learning on out-of-fold predictions

- PyTorch, HuggingFace Transformers, TensorFlow/Keras, scikit-learn, ONNX Runtime

NLP, LLMs & generative AI

- LLMs – prompt and context engineering, RAG, vector databases, fine-tuning, tool use and agentic workflows
- Classical and supervised NLP – tokenization, topic models, text classification, sentiment and stance detection, NER, POS tagging, dependency parsing
- spaCy, NLTK, gensim, Prodigy

Speech & language technology

- Phoneme-level ASR – wav2vec2 XLS-R, CTC, bespoke IPA tokenizers
- Autoregressive encoder-decoder ASR and CTC/AED ensembling
- Goodness-of-pronunciation estimation for language-impaired children
- Diarization; text-to-speech for clinical and educational settings

Statistics & experimental design

- Bayesian and mixed-effects models; generalized linear models; ANOVA
- Causal DAGs and mediation analysis; power analysis
- Lab and online experiments, randomized controlled trials, psycholinguistic paradigms
- Psychometric scale construction and validation; survey design with attention checks
- R – lme4, brms, tidyverse, ggplot2, sjPlot, caret

Systems & infrastructure

- Python, TypeScript, SQL, R
- AWS (SageMaker, Lambda, ECS, S3); Azure OpenAI
- React Native; Supabase; Django, Flask, Dash
- Docker; Git and GitHub; CI/CD with GitHub Actions
- Config-driven, seed-controlled experiment pipelines with run ledgers
- Large text and audio datasets; object storage

OTHER

- I am a believer in using AI for good, and volunteered for two projects with the Rhodes AI Lab, which solves social problems using AI: Toxic comments – worked with Google engineers to classify toxic online comments; Future of work – used NLP to track effects of automation on labor markets.

- Led research on Texture Generate, which combines a deep auto-ML ensemble, xAI techniques, and generative AI to improve government and third-sector communications performance: a stacked ensemble of a fine-tuned transformer and gradient boosting over 19 curated psycholinguistic word norms, with attributions fed to a generative copywriter. Live A/B tests through media agencies showed 20–50% gains in click-through.
- Moved the product from stochastic surrogate explanations to deterministic gradient attribution after comparing explainers on rerun stability and downstream A/B performance.
- Led research and engineering on Pulse, a social listening platform based on adapted Latent Dirichlet Allocation techniques.
- Designed and built a government communication optimisation service, Target. Target improved click-through rates for a public-service campaign by 27%.

GRANTS AND FUNDING

Awarded

- **Innovate UK SMART Grant** (£200k, 2021). Lead author. Texture Target: A software service predicting and responding to COVID-19 outbreaks in UK regions using Twitter data. Applied xAI, NLP, and my PhD research to improve government communications around the pandemic; led collaborations with Cardiff and Lancaster Universities and worked with commercialization experts to build the service.
- **NSERC Postgraduate Research Fellowship (Doctoral)**, Canada (£65k, 2018). Nationally competitive; funded an independent transdisciplinary programme blending economics, linguistics, and cognitive science, including a deterministic multilingual future-tense text classification system.
- **Visiting Doctoral Research Fellowship**, Department of Experimental Psychology, University of Oxford (£90k, 2015).
- **Declined:** Doctoral Fellowship, University of Bristol (£90k, 2015); University Fellowship (Doctoral), George Washington University (£80k, 2015); Doctoral Fellowship, Department of Anthropology, University of Oxford (£30k, 2014).

Under review

- **Schmidt Sciences**, Scaling AI Safety for a Multi-Agent World (Tier 1, US\$300k, 24 months; submitted 2026). Named Research Associate at 100% FTE; PI Soumya Banerjee, University of York. Emergent goals in populations of generalizing agents: detecting collective goal drift as a multi-agent safety risk.
- **John Templeton Foundation**, Intelligence Venture (submitted 2026, with Soumya Banerjee). Named Research Associate at 100% FTE, Winter 2027. Whether fluid-intelligence benchmark performance generalizes to world modeling.

AI SAFETY FELLOWSHIP APPLICATIONS (PENDING)

- Anthropic AI Safety Fellows Program (reached final round, decision pending).
- Santa Fe Institute Complexity Postdoctoral Fellowship (2027).

RESEARCH SOFTWARE AND OPEN ARTEFACTS

- **mental_models** — experimental harness for the LLM mechanical-reasoning studies: procedural generation of pulley systems in TikZ with causally relevant and irrelevant features crossed, a config-driven multi-model trial runner, and analysis.
- **OgmaML** — reproducible training and evaluation pipeline for phoneme-level speech recognition on disordered child speech: ingest for CHILDES, PhonBank and KidLUCID, quality filtering, speaker-respecting splits, and seed-controlled runs logged to a ledger (~14,000 lines, 21 test modules). An open, reproducible dataset release is in preparation with Brian MacWhinney (TalkBank, Carnegie Mellon).
- **amyloid-app.com** — interactive 3-D view of the npj Dementia classifier’s representation space, tokens scaled by attribution weight.

AWARDS AND HONOURS

- Nominated, Glushko Dissertation Prize, Cognitive Science Society.
- Shortlisted, Tech for Good, Campaign Magazine (2023), for Texture Target.
- Nicola Knight Prize for Best Quantitative Methods in an MSc, University of Oxford (2014).

SELECTED TALKS AND PRESENTATIONS

- **Robertson, C., & Wolff, P.** (2025). Cognitive-science approaches to evaluating mechanical reasoning in LLMs: Output layer evidence of shallow world-model use [Accepted talk]. Superintelligence Conference (SiC).
- **Robertson, C., & Wolff, P.** (2025). LLMs have “mental” models: Latent world models in LLM network weights can be inferred from output layer inference [Accepted poster]. Cognitive Science.
- **Robertson, C.** (2024). How would industry like to engage with universities to develop AI. Panel discussion, Centre for Human-Inspired AI, Cambridge.
- **Robertson, C.** (2023). Using cognitive science methods to probe latent representations in large language models. Working paper series, Emory University.

- **Robertson, C.** (2020). Not when but if: Probability not time determines cross-linguistically driven differences in psychological discounting. *Cognitive Linguistics*.

TEACHING AND SUPERVISION

- **Supervision.** Undergraduate honors dissertations at Emory University (2024; graduated with highest honors, work published in *Cognition*) and the London Interdisciplinary School (2023; graduated with honors, annual best-dissertation award).
- **Teaching.** Lecturer, London Interdisciplinary School (2026): statistics, research methods, data science and applied machine learning. Lecturer, CBL Academies (2017–2019): academic writing, critical thinking, statistics, applied machine learning. Tutor, Worcester College (2017–2018): psychology, methods for behavioral sciences, data analysis in R.

IN THE PRESS

- Psychology Today: “Does language shape how we think about the future?”
- Long form interview, Because Language (YouTube): “Back to the FTR”; Research Featured, Because Language (YouTube): “Learning from LLMs”.

SELECTED PUBLICATIONS

In preparation

- Robertson, C.,** & Wolff, P. (In prep). Functional localization of emergent world-model reasoning in frontier large language models.
- Robertson, C.,** & Dunbar, R. I. M. (In prep). Linguistic recursion and the depth of embedded theory of mind in humans and large language models.
- Robertson, C.** (In prep). Ecologies of human–machine interdependence: Selecting for cross-boundary mutualism in multi-agent systems.
- Robertson, C.,** Tomanek, K., & Hain, T. (In prep). A novel transfer-learning approach for connectionist temporal classification to estimate goodness of pronunciation in language-impaired children.

Published

- Robertson, C.,** Hochberg, D., Quimby, M., Wolff, P., Dickerson, B. C., & Rezaii, N. (2026). Predicting amyloid status in primary progressive aphasia using explainable artificial intelligence. *npj Dementia*, 2, Article 41. <https://doi.org/10.1038/s44400-026-00092-w>
- Robertson, C.,** Roberts, S. G., Majid, A., Lu, T., Wolff, P., & Dunbar, R. I. M. (2026). Low-certainty modals not future tenses cause increased psychological discounting in English relative to Dutch. *Cognition*, 267, Article 106338. <https://doi.org/10.1016/j.cognition.2025.106338>
- Robertson, C.,** & Wolff, P. (2025). LLM world models are mental: Output layer evidence of brittle world model use in LLM mechanical reasoning. In *Proceedings of the Superintelligence Conference (SiC)*. Full paper: [arXiv:2507.15521](https://arxiv.org/abs/2507.15521)
- Robertson, C.,** Roberts, S. G., Majid, A., & Dunbar, R. I. M. (2025). Language and economic behaviour: Future tense use causes less not more temporal discounting. *PLoS ONE*, 20(5), e0317422. <https://doi.org/10.1371/journal.pone.0317422>
- Robertson, C.,** & Roberts, S. G. (2023). Not when but whether: Modality and future time reference in English and Dutch. *Cognitive Science*, 47(1), e13224. <https://doi.org/10.1111/cogs.13224>
- Robertson, C.,** Carney, J., & Trudell, S. (2023). Language about the future on social media as a novel marker of anxiety and depression: A big-data and experimental analysis. *Current Research in Behavioral Sciences*, 4, 100104. <https://doi.org/10.1016/j.crbeha.2023.100104>
- Carney, J., & **Robertson, C.** (2022). Five studies evaluating the impact on mental health and mood of recalling, reading, and discussing fiction. *PLOS ONE*, 17(4), e0266323. <https://doi.org/10.1371/journal.pone.0266323>
- Gotti, G., Roberts, S. G., Fasan, M., & **Robertson, C.** (2021). Language in economics and accounting research: The role of linguistic history. *The International Journal of Accounting*, 56(3), 2150015. <https://doi.org/10.1142/S1094406021500153>
- Robertson, C.** (2021). *The future is a thing of possibility: Cross-linguistic differences in modal future time reference affect (risky) intertemporal decisions*. Oxford: Holywell Press Limited.
- Carney, J., **Robertson, C.,** & Dávid-Barrett, T. (2019). Fictional narrative as a variational Bayesian method for estimating social dispositions in large groups. *Journal of Mathematical Psychology*, 93, 102279. <https://doi.org/10.1016/j.jmp.2019.102279>
- Carney, J., & **Robertson, C.** (2018). People searching for meaning in their lives find literature more engaging. *Review of General Psychology*, 22(2), 199–209. <https://doi.org/10.1037/gpr0000134>
- Dunbar, R. I. M., MacCarron, P., & **Robertson, C.** (2018). Trade-off between fertility and predation risk drives a geometric sequence in the pattern of group sizes in baboons. *Biology Letters*, 14(3), 20170700. <https://doi.org/10.1098/rsbl.2017.0700>
- Robertson, C.,** Tarr, B., Kempnich, M., & Dunbar, R. (2017). Rapid partner switching may facilitate increased broadcast group size in dance compared with conversation groups. *Ethology*, 123(10), 736–747. <https://doi.org/10.1111/eth.12642>
- Dunbar, R. I. M., Launay, J., Włodarski, R., **Robertson, C.,** Pearce, E., Carney, J., ... (2017). Functional benefits of (modest) alcohol consumption. *Adaptive Human Behavior and Physiology*, 3(2), 118–133. <https://doi.org/10.1007/s40750-016-0058-4>
- Dunbar, R., Launay, J., Pearce, E., Włodarski, R., Carney, J., MacCarron, P., **Robertson, C.** (2017). *Friends on tap: The role of pubs at the heart of the community*. St. Albans, UK: Campaign for Real Ale.